

BANJA LUKA INTL, Bosnia and Herzegovina

WMO: 132420

Lat: 44.941N

Lon: 17.298E

Elev: 400

StdP: 14.48

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.2	17.7	7.2	8.1	14.0	13.7	11.2	19.5	16.0	42.7	13.8	43.0	2.8	160	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	94.7	71.2	91.2	71.1	87.8	69.8	74.2	87.5	72.7	86.2	71.1	84.0	5.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.1	112.3	80.0	68.3	105.5	78.3	66.6	99.2	76.2	37.9	87.3	36.5	86.2	35.3	84.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.0	10.0	DB	4.8	98.7	8.1	4.2	-1.0	101.7	-5.8	104.2	-10.3	106.5	-16.2	109.5	
			WB	4.2	77.5	7.9	1.6	-1.5	78.7	-6.1	79.6	-10.6	80.5	-16.3	81.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	34.2	36.9	45.5	55.0	61.8	69.3	72.9	71.8	63.3	54.1	45.2	36.1
	DBStd	15.42	8.94	9.21	8.58	6.24	6.51	6.54	5.56	5.76	6.59	7.07	7.80	8.21
	HDD50	1733	492	371	184	26	4	0	0	0	39	180	437	(8)
	HDD65	4785	956	787	605	304	139	33	6	12	109	344	594	896
	CDD50	3159	2	3	44	175	368	578	708	676	398	165	36	6
	CDD65	738	0	0	0	3	39	162	250	222	57	5	0	0
	CDH74	8322	0	0	4	106	526	1741	2781	2415	665	82	2	0
	CDH80	3472	0	0	0	17	126	683	1289	1125	225	7	0	0
(14) Wind	WSAvg	4.0	3.7	4.1	4.8	4.5	4.4	4.2	4.1	3.9	3.8	3.5	3.5	3.7
(15) Precipitation	PrecAvg	41.9	2.8	2.6	3.2	3.7	4.2	4.2	3.4	3.5	4.2	3.5	3.7	3.5
	PrecMax	63.7	5.4	4.6	5.9	8.5	8.1	8.1	8.6	12.5	10.4	6.9	8.0	7.0
	PrecMin	24.8	0.5	0.6	0.4	0.7	1.8	2.2	1.4	0.4	1.2	0.4	0.4	0.2
	PrecStd	8.5	1.3	1.1	1.3	1.9	2.0	1.6	1.7	3.0	2.6	1.9	1.8	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	65.8	73.6	82.5	87.8	93.5	98.7	99.0	91.7	80.5	71.9	64.6
		MCWB	51.2	51.7	54.7	62.5	68.4	73.0	71.4	70.2	68.7	64.5	58.9	54.2
	2%	DB	55.7	59.3	69.7	76.8	83.9	90.0	94.8	94.7	86.1	76.6	66.5	59.1
		MCWB	48.2	48.9	53.5	59.5	67.4	71.7	71.5	70.4	66.9	62.4	56.1	51.1
	5%	DB	50.3	55.2	64.8	73.2	80.3	87.5	91.3	90.0	81.0	71.7	62.4	53.7
		MCWB	45.8	47.0	51.8	57.8	64.3	70.5	71.6	70.3	66.0	61.0	54.0	47.6
	10%	DB	46.0	50.0	60.8	69.5	76.6	84.2	87.7	86.3	77.1	66.4	57.2	47.9
		MCWB	43.1	45.1	49.9	55.9	63.3	68.9	69.8	69.4	64.3	58.5	52.0	45.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.7	52.5	57.3	63.7	70.2	75.9	76.9	75.4	71.2	66.0	60.1	55.1
		MCDB	59.5	62.7	69.1	78.9	83.9	88.7	89.7	87.9	82.9	77.3	69.5	63.7
	2%	WB	48.9	50.0	54.7	60.8	68.2	73.8	74.3	73.4	68.4	63.6	57.3	51.2
		MCDB	54.9	58.0	66.0	73.0	80.8	86.7	87.7	87.0	81.8	73.1	65.2	58.4
	5%	WB	46.0	47.6	52.6	58.9	66.2	71.9	72.6	72.0	66.9	61.5	55.0	48.2
		MCDB	49.8	54.2	63.2	69.2	77.2	83.7	86.8	85.6	78.9	70.0	61.2	53.5
	10%	WB	42.8	45.0	50.6	57.2	64.0	69.9	71.1	70.4	65.4	59.5	52.4	44.7
		MCDB	44.7	48.7	59.5	66.7	73.8	81.4	84.2	83.1	74.8	66.2	57.3	46.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	14.8	19.3	21.9	20.8	21.1	22.9	23.3	20.8	19.1	15.6	12.2
		MCDBR	20.9	25.1	28.8	30.8	27.9	27.1	29.9	31.4	28.7	27.6	25.2	22.4
	5% WB	MCWBR	14.6	15.6	15.4	15.2	13.0	11.5	10.9	11.3	12.7	15.4	15.7	15.3
		MCDWR	20.0	22.8	24.9	25.4	25.0	24.7	25.8	26.3	25.0	24.8	23.1	20.5
(40) Clear-Sky Solar Irradiance	taub	0.332	0.361	0.395	0.448	0.436	0.435	0.435	0.435	0.436	0.421	0.390	0.353	0.332
		taud	2.388	2.303	2.247	2.118	2.202	2.247	2.243	2.249	2.264	2.346	2.402	2.402
	Ebn at Noon	245	257	263	258	266	266	264	259	252	243	234	231	(42)
		Edn at Noon	24	31	38	47	44	43	43	41	37	30	24	22
(44) All-Sky Solar Radiation	RadAvg	404	618	996	1358	1683	1904	1939	1726	1181	802	467	334	(44)
	RadStd	59	92	136	163	148	158	137	191	147	95	57	58	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	+0.90	N/A	N/A	N/A	+0.63	N/A	N/A	-216	N/A	N/A